Tarea Final

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1. Para el proceso $ARMA(1,2)$, calcule $\gamma_0, \gamma_1, \gamma_2, \gamma_k, k > 2$. Exprese $\gamma_0, \gamma_1, \gamma_2, \gamma_k$ en términos de los parámetros autorregresivos, de media móvil y de σ_Z^2.

Consideramos Modelo ARMA(1,2)

$$(1-\phi B)X_t = (1-\theta_1 B - \theta B^2) Z_t$$

$$X_t - \phi X_{t-1} = Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2}$$

$\gamma_0 = E[X_t X_t]$ De esta manera:

$$E[X_t X_t] = E[\phi_1 X_{t-1} X_t + Z_t X_t - \theta_1 Z_{t-1} X_t - \theta_2 Z_{t-2} X_t]$$

$$\gamma_0 = \phi_1 E[X_{t-1} X_t] + E[X_t Z_t] - \theta_1 E[Z_{t-1} X_t] - \theta_2 E[Z_{t-2} X_t]$$

- $\phi_1 E[X_{t-1} X_t] = \phi_1 \gamma_1$
- $E[Z_t X_t] = E[\phi Z_t X_{t-1} + Z_t Z_t - \theta_1 Z_{t-1} Z_t - \theta_2 Z_{t-2} Z_t]$
    - $\phi E[Z_t X_{t-1}]$ $i=0$ $k=1$ $\forall\, i < k = 0$
    - $E[Z_t^2] = \sigma_z^2$
    - $E[Z_{t-1} Z_t] = 0$
    - $E[Z_{t-2} Z_t] = 0$
    - $= \sigma_z^2$
- $-\theta_1 E[Z_{t-1} X_t] = \phi\, E[Z_{t-1} X_{t-1}] - \theta_1 \sigma_z^2$
    $= \phi \sigma_z^2 - \theta_1 \sigma_z^2$
    $= -\theta_1 (\phi \sigma_z^2 - \theta_1 \sigma_z^2)$
    $= -\theta_1 \sigma_z^2 (\phi - \theta_1)$
- $-\theta_2 E[Z_{t-2} X_t] = \phi_1 E[Z_{t-2} X_{t-1}] - \theta_2 \sigma_z^2$
    $= \phi_1 \sigma_z^2 (\phi_1 - \theta_1) - \theta_2 \sigma_z^2$
    $= -\theta_2 \sigma_z^2 (\phi^2 - \theta_1 \phi_1 - \theta_2)$

$$\gamma_0 = \phi_1 \gamma_1 + \sigma_z^2 - \theta_1 (\phi_1 \sigma_z^2 - \theta_1 \sigma_z^2) - \theta_2 (\phi^2 \sigma_z^2 - \theta_1 \phi_1 \sigma_z^2 - \theta_2 \sigma_z^2)$$

$$\gamma_0 = \phi \gamma_1 \;\; \sigma_z^2 [1 - \theta_1 \phi_1 + \theta_1^2 - \theta_2 \phi^2 + \theta_1 \theta_2 \phi_1 + \theta_2^2]$$

$\gamma_1 = E[X_t X_{t-1}] = E[\phi_1 X_{t-1} X_{t-1} + X_{t-1} Z_t - \theta_1 Z_{t-1} X_{t-1} - \theta_2 X_{t-1} Z_{t-2}]$

$= \phi E[X_{t-1} X_{t-1}] + E[X_{t-1} Z_t] - \theta_1 E[X_{t-1} Z_{t-1}] - \theta_2 E[X_{t-1} Z_{t-2}]$

$= \phi \gamma_0 + 0 - \theta_1 \sigma_z^2 - \theta_2 (\sigma_z^2 \phi - \theta_1 \sigma_z^2)$

$\gamma_1 = \phi \gamma_0 + \sigma_z^2 [-\theta_1 - \theta_2 \phi + \theta_1 \theta_2]$

$\gamma_2 = E[X_t X_{t-2}] = E[\phi X_{t-1} X_{t-2} + Z_t X_{t-2} - \theta_1 Z_{t-1} X_{t-2} - \theta_2 Z_{t-2} X_{t-2}]$

$= \phi E[X_{t-1} X_{t-2}] + E[Z_t X_{t-2}] - \theta_1 E[Z_{t-1} X_{t-2}] - \theta_2 E[Z_{t-2} X_{t-2}]$

$= \phi \gamma_1 + 0 + 0 - \theta_2 \sigma_z^2$

$= \phi \gamma_1 - \theta_2 \sigma_z^2$

$\gamma_3 = E[X_t X_{t-3}] = E[\phi X_{t-1} X_{t-3} + Z_t X_{t-3} - \theta_1 Z_{t-1} X_{t-3} - \theta_2 Z_{t-2} X_{t-3}]$

[illegible]

$$\gamma_3 = E[X_t X_{t-3}] = E[\phi X_{t-1} X_{t-3} + Z_t X_{t-3} - \theta_1 Z_{t-1} X_{t-3} - \theta_2 Z_{t-2} X_{t-3}]$$
$$= \phi E[X_{t-1} X_{t-3}] + E[X_{t-3} Z_t] - \theta_1 E[Z_{t-1} X_{t-3}] - \theta_2 E[Z_{t-2} X_{t-3}]$$
$$= \phi \gamma_2 + 0 - 0 - 0$$
$$= \phi \gamma_2$$

$$\gamma_k = \phi \gamma_{k-1} \quad \forall k > 2$$

$$\gamma_k \begin{cases} \phi \gamma_1 + \sigma_z^2 [1 - \phi\theta_1 + \theta_1^2 - \theta_2\phi^2 + \theta_1\theta_2\phi + \theta_2^2] & , |k| = 0 \\ \phi_1 \gamma_0 + \sigma_z^2 [-\theta_1 - \theta_2 \phi + \theta_1\theta_2] & , |k| = 1 \\ \phi_1 \gamma_1 - \theta_2 \sigma_z^2 & , |k| = 2 \\ \phi_1 \gamma_{k-1} & , |k| > 2 \end{cases}$$

Para futuros ejercicios γ_0 despejado:

$$\gamma_0 = \phi(\phi_1 \gamma_0 + \sigma_z^2 [-\theta_1 - \theta_2\phi + \theta_1\theta_2]) + \sigma_z^2 [1 - \phi\theta_1 + \theta_1^2 - \theta_2\phi^2 + \theta_1\theta_2\phi + \theta_2]$$
$$\gamma_0 = \phi_1^2 \gamma_0 + \sigma_z^2 [-\phi\theta_1 - \theta_2\phi^2 + \theta_1\theta_2\phi + 1 - \phi\theta_1 + \theta_1^2 - \theta_2\phi^2 + \theta_1\theta_2\phi + \theta_2^2]$$
$$(1-\phi)\gamma_0 = \sigma_z^2 [1 - 2\phi\theta_1 - 2\theta_2\phi^2 + 2\theta_1\theta_2\phi + \theta_1^2 + \theta_2^2]$$

$$\gamma_0 = \frac{1 - 2\phi\theta_1 - 2\theta_2\phi^2 + 2\theta_1\theta_2\phi + \theta_1^2 + \theta_2^2}{(1-\phi^2)} \sigma_z^2$$

2. Considere los siguientes modelos

I. $\tilde{X}_t - 0.9\tilde{X}_{t-1} = Z_t + 0.8Z_{t-1}$

II. $\tilde{X}_t + 0.1\tilde{X}_{t-1} = Z_t + 0.8Z_{t-1}$

III. $\tilde{X}_t + 0.1\tilde{X}_{t-1} = Z_t + 0.2Z_{t-1} - 0.7Z_{t-2}$

En donde $\{Z_t\}$ es un proceso de ruido blanco con media cero y varianza constante.

a) Verifique si los procesos son estacionarios e invertibles. Identifique la región admisible de cada proceso (solo para el proceso I y II).
b) Calcule las 5 primeras autocorrelaciones simples teóricas de cada proceso.
c) Calcule las 3 primeras autocorrelaciones parciales teóricas de cada proceso.

Para este ejercicio considerar para ARMA(1,1)

$$\rho_k = \frac{\theta^{k-1}(1-\phi\theta)(\phi-\theta)}{1 - 2\phi\theta + \theta^2} \qquad \phi_{11} = \rho_1 \qquad \phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

$$\phi_{33} = \frac{\det \begin{vmatrix} 1 & \rho_1 & \rho_1 \\ \rho_1 & 1 & \rho_2 \\ \rho_2 & \rho_1 & \rho_3 \end{vmatrix}}{\det \begin{vmatrix} 1 & \rho_1 & \rho_2 \\ \rho_1 & 1 & \rho_1 \\ \rho_2 & \rho_1 & 1 \end{vmatrix}}$$

① $X_t - 0.9X_{t-1} = Z_t + 0.8Z_{t-1}$

$\phi = 0.9$ $\quad |\phi| < 1$ $\quad$ Es estacionario e invertible

$\theta = -0.8$ $\quad |\theta| < 1$ $\quad$ Región admisible C

$\phi = 0.9 \quad |\phi|<1$ Es estacionario e invertible

$\theta = -0.8 \quad |\theta|<1$ Región admisible 6

Primeras 5 ρs

$\rho_1 = 0.949351$

$\rho_2 = 0.854416$

$\rho_3 = 0.768974$

$\rho_4 = 0.692077$

$\rho_5 = 0.622869$

Primeras 3 ϕ_{pp}

$\phi_{11} = 0.949351$

$\phi_{22} = -0.474521$

$\phi_{33} = \frac{0.0023\,1109}{0.0075535} = 0.30596$

② $X_t + 0.1X_{t-1} = Z_t + 0.8\,Z_{t-1}$

$\phi = -0.1 \quad |\phi|<1$ Es invertible y estacionario

$\theta = -0.8 \quad |\theta|<1$ Región Admisible 5

Primeras 5 ρs

$\rho_1 = 0.4351$

$\rho_2 = -0.04351$

$\rho_3 = 0.004351$

$\rho_4 = -0.0004351$

$\rho_5 = 0.00004351$

Primeras 3 ϕ_{pp}

$\phi_{11} = 0.4351$

$\phi_{22} = -.23888$

$\phi_{33} = \frac{0.12458}{0.603008} = 0.20660$

③ ARMA(1,2)

$X_t + 0.1X_{t-1} = Z_t + 0.2Z_{t-1} - 0.7Z_{t-2}$

$\phi = -0.1 \quad |\phi|<1$ Estacionario

Polinomio $\quad 0 = 1 + 0.2x - 0.7x^2$

raíces $\quad x_1 = -1.060878 \quad |z|>1$

$x_2 = 1.346593 \quad |z|>1$

Primeras 5 ρs $\quad \phi = -0.1 \quad \theta_1 = -0.2 \quad \theta_2 = 0.7$

$$r_0 = \frac{1 - 2\phi\theta_1 - 2\theta_2\phi^2 + 2\theta_1\theta_2\phi + \theta_1^2 + \theta_2^2}{(1-\phi^2)}\sigma_z^2 = 1.51\bar{9}$$

$\rho_1 = -0.0219/1.51\bar{9} = -0.014$

$\rho_2 = -0.6978/1.5199 = -0.45$

$\rho_3 = 0.045$

$\rho_4 = -0.0045$

$\rho_5 = 0.00045$

Primeras 3 ϕ_{pp}s

$\phi_{11} = -0.014$

$\phi_{22} = -0.45028$

$\phi_{33} = 0.050$

3. Los procesos del ejercicio anterior pueden expresarse también como $\pi(B)\tilde{X}_t = Z_t$ y como $\tilde{X}_t = \psi(B)Z_t$. Para cada uno de los modelos, encuentre $\pi_1, \dots, \pi_5$ y $\psi_1, \dots, \psi_5$.

$X_t = (\theta)Z_t$

$\downarrow$

∞

$X_t = 1 - \sum_{i=1}^{\infty} \psi_i \, (Z_t - u)$

$\theta(\phi)^{-1}$

Ya que se mostró que todos los procesos son invertibles

$\hat{\phi}(B)\hat{X}_t = \hat{\theta}(B)Z_t$

$\hat{X}_t = [\hat{\phi}B]^{-1}\hat{\theta}(B)Z_t$

$\hat{X}_t = \hat{\psi}(B)Z_t$

$\hat{X}_t = (1 - \hat{\psi}_1 B - \hat{\psi}_2 B^2 - \hat{\psi}_3 B^3 - \dots)Z_t$

$\hat{X}_t = Z_t - \hat{\psi}_1 Z_{t-1} - \hat{\psi}_2 Z_{t-2} - \dots$

$Z_t = \hat{X}_t - \hat{\psi} Z_{t-1} - \hat{\psi}_2 Z_{t-2} \dots$

$\hat{Z}_t = \hat{\psi}_1 Z_{t-1} - \hat{\psi}_2 Z_{t-2} \dots$

$\hat{X}_t = Z_t - \hat{Z}_t$

De manera similar se puede expresar Z_t como un polinomio infinito de $\hat{\pi}$s

$\hat{Z}_t = X_t - \hat{X}_t$

Notar que se llega a

$\phi(B)\psi(B) = \theta(B)$

$\theta(B)\pi(B) = \phi(B)$

Para encontrar πs de ARMA(1,1)

$\theta(B)\pi(B) = \phi(B)$ $\quad$ $\pi(B)$ es un polinomio inf.

$1 - \phi B = (1 - \theta B)\pi(B)$

$= \pi(B) - \theta B \pi(B)$

$= 1 - \pi_1 B - \pi_2 B^2 - \pi_3 B^3 - \dots - \theta B + \theta B^2 \pi_1 + \theta B^3 \pi_2 + \theta B^4 \pi_3 \dots$

$= 1 - B(\pi_1 + \theta) - B^2(\pi_2 - \theta\pi_1) - B^3(\pi_3 - \theta\pi_2) \dots$

Si tomamos solo los binomios con B^1, y consideramos los demas $= 0$

$1 - \phi B = 1 - B(\pi_1 + \theta)$

$\phi = \pi_1 + \theta$

- $\pi_1 = \phi - \theta$
- $\pi_2 = \theta\pi_1$
- $\pi_3 = \theta\pi_2$
- $\pi_4 = \theta\pi_3$
- $\pi_5 = \theta\pi_4$

Para encontrar ψ's para ARMA(1,1)

$\phi(B)\psi(B) = \theta(B)$ $\quad$ donde ψ es un polinomio inf.

$1 - \theta B = (1 - \phi B)\psi(B)$

$= \psi(B) - \phi B \psi(B)$

$= 1 - \psi_1 B - \psi_2 B^2 - \psi_3 B^3 - \phi B + \phi B^2 \psi_1 + \phi B^3 \psi_2 + \phi B^4 \psi_3 \dots$

$= 1 - B(\psi_1 - \phi) - B^2(\psi_2 - \psi_1\phi) - B^3(\psi_3 - \psi_2\phi) \dots$

Igualamos todos los B a 0 menos el primer binomio

$\theta = \psi_1 - \phi$

$\psi_1 = \theta - \phi$

$\psi_2 = \psi_1\phi$

$\psi_3 = \psi_2\phi$

$\psi_2 = \psi_1 \phi$

$\psi_3 = \psi_2 \phi$

$\psi_4 = \psi_3 \phi$

$\psi_5 = \psi_4 \phi$

Para encontrar π_j de ARMA(1,2)

$\phi(B) = \theta(B)\pi(B)$

$1 - \phi B = (1-\theta_1 B - \theta_2 B^2)\,\pi(B)$

$= \pi(B) - \theta_1 B \pi(B) - \theta_2 B^2 \pi(B$

$= 1 - \pi_1 B - \pi_2 B^2 - \pi_3 B^3 \ldots - \theta_1 B + \theta_1 B^2 \pi_1 + \theta_1 B^3 \pi_2 + \theta_1 B^4 \pi_3 \ldots - \theta_2 B^2 + \theta_2 B^3 \pi_1 + \theta_2 B^4 \pi_2$

$= 1 - B(\pi_1 - \theta_1) - B^2(\pi_2 - \theta_1\pi_1 - \theta_2) - B^3(\pi_3 - \theta_1\pi_2 - \theta_2\pi_1) - \ldots$

misma tecnica igualamos a 0

$\pi_1 = \phi - \theta$

$\pi_2 = \theta_1 \pi_1 + \theta_2$

$\pi_3 = \theta_1 \pi_2 + \theta_2 \pi_1$

$\pi_4 = \theta_1 \pi_3 + \theta_2 \pi_2$

$\pi_5 = \theta_1 \pi_4 + \theta_2 \pi_3$

Para encontrar ψ_s de ARMA(1,2)

$(\theta)B = (\phi)B(\psi)B$

$1 - \theta_1 B - \theta_2 B^2 = (1 - \phi B)(\psi)B$

$= (\psi)B - \phi B(\psi)B$

$= 1 - B\psi_1 - \psi_2 B^2 - \psi_3 B^3 - \ldots - \phi B + \phi B^2 \psi_1 + \phi B^3 \psi_2 + \ldots$

$= 1 - B(\psi_1 + \phi) - B^2(\psi_2 - \phi\psi_1) - B^3(\psi_3 - \phi\psi_2) - \ldots$

$\theta_1 B = B(\psi_1 + \phi)$

$\psi_1 = \theta_1 - \phi$

$\theta_2 B^2 = B^2(\psi_2 - \phi\psi_1)$

$\psi_2 = \theta_2 + \phi\psi_1$

$\psi_3 = \phi\psi_2$

$\psi_4 = \phi\psi_3$

$\psi_5 = \phi\psi_4$

1) $X_t - 0.9X_{t-1} = Z_t + 0.8Z_{t-1}$ $\quad \phi = 0.9 \quad \theta = -0.8$

- $\pi_1 = \phi - \theta = -1.7$ $\qquad \psi_1 = \theta - \phi = -1.7$
- $\pi_2 = \theta\pi_1 = 1.36$ $\qquad \psi_2 = \psi_1\phi = -1.53$
- $\pi_3 = \theta\pi_2 = -1.088$ $\qquad \psi_3 = \psi_2\phi = -1.377$
- $\pi_4 = \theta\pi_3 = 0.8704$ $\qquad \psi_4 = \psi_3\phi = -1.2393$
- $\pi_5 = \theta\pi_4 = -0.696$ $\qquad \psi_5 = \psi_4\phi = -1.11537$

② $X_t + 0.1X_{t-1} = Z_t + 0.8Z_{t-1}$ $\phi = -0.1$ $\theta = -0.8$

- $\pi_1 = \phi - \theta = 0.7$ $\psi_1 = \theta - \phi = -0.7$
- $\pi_2 = \theta\pi_1 = .56$ $\psi_2 = \psi_1\phi = 0.07$
- $\pi_3 = \theta\pi_2 = .448$ $\psi_3 = \psi_2\phi = -0.007$
- $\pi_4 = \theta\pi_3 = .3584$ $\psi_4 = \psi_3\phi = 0.0007$
- $\pi_5 = \theta\pi_4 = .2872$ $\psi_5 = \psi_4\phi = -0.00007$

③ $X_t + 0.1X_{t-1} = Z_t + 0.2Z_{t-1} - 0.7Z_{t-2}$ $\phi = -0.1$ $\theta_1 = -0.2$ $\theta_2 = 0.7$

- $\pi_1 = \phi - \theta_1 = -0.3$ $\psi_1 = \theta_1 - \phi = 0.3$
- $\pi_2 = \theta_1\pi_1 + \theta_2 = 0.64$ $\psi_2 = \theta_2 + \phi\psi_1 = 0.67$
- $\pi_3 = \theta_1\pi_2 + \theta_2\pi_1 = -0.082$ $\psi_3 = \phi\psi_2 = -0.067$
- $\pi_4 = \theta_1\pi_3 + \theta_2\pi_2 = 0.4316$ $\psi_4 = \phi\psi_3 = 0.0067$
- $\pi_5 = \theta_1\pi_4 + \theta_2\pi_3 = 0.0289$ $\psi_5 = \phi\psi_4 = -0.00067$

4. Considere el proceso ARIMA(1,1,2) descrito por

$$(1-0.4B)\nabla Z_t = (1+0.5B-0.14B^2)a_t$$

Donde a_t es un ruido blanco con media cero y varianza constante.

Diga si cada uno de los siguientes procesos es estacionario y/o invertible. Justifique su respuesta.

a) Z_t
b) ∇Z_t
c) $\nabla^2 Z_t$
d) $\frac{1}{2}\nabla Z_t + \frac{1}{2}\nabla Z_{t-1}$

a) Z_t

$(1-0.4B)\nabla Z_t = (1+0.5B-0.14B^2)a_t$

$(1-0.4B)(1-B)Z_t = (1+0.5B-0.14B^2)a_t$

$(1-B-0.4B+0.4B^2)Z_t = (1+0.5B-0.14B^2)a_t$

$(1-1.4B+0.4B^2)Z_t = (1+0.5B-0.14B^2)a_t$

Estacionariedad ∴ No $|\phi| > 1$

$0.4 - 1.4 > 1$

Invertibilidad

- $\theta(B)$ es igual al pasado

Es invertible

b) ∇Z_t $(1-0.4B)\nabla Z_t = (1+0.5B-0.14B^2)a_t$

Estacionariedad

$|\phi| < 1$

Estacionariedad

$|\phi| < 1$

$|0.4| < 1$

Si es

Invertibilidad

$0.14 - 0.5 < 1$

$0.14 + 0.5 < 1$

$|0.5| < 1$

Estacionario e invertible

c) $\nabla^2 Z_t$

$$(1-0.4B)\nabla^2 Z_t = (1 + 0.5B - 0.14B^2)\nabla \alpha_t$$

Estacionario

$|0.4| < 1$

Es estacionario.

Tiene una raiz $= 1$

No es invertible

d) $\frac{1}{2}\nabla Z_t + \frac{1}{2}\nabla Z_{t-1}$

$$\frac{1}{2}\nabla Z_t + \frac{1}{2}\nabla Z_{t-1} = \frac{1}{2}(Z_t - Z_{t-1}) + \frac{1}{2}(Z_{t-1} - Z_{t-2})$$

$$= \frac{1}{2}(Z_t - Z_{t-1})$$

$$= \frac{1}{2}(1-B^2)Z_t = \frac{1}{2}(1-B)(1+B)$$

$$= \frac{1}{2}(1+B)\nabla Z_t$$

$$\frac{1}{2}(1+B)(1-0.4B)\nabla Z_t = \frac{1}{2}(1+B)(1+0.5B-0.14B^2)\alpha_t$$

$$(1-0.4B)\left(\frac{1}{2}\nabla Z_t + \frac{1}{2} Z_{t-2}\right) = \frac{1}{2}(1+B)(1+0.5B-0.14B^2)\alpha_t$$

Estacionario

$|0.4| < 1$

Es estacionario.

No invertible tiene

unar raiz $= 1$